ENGINEERING TRAINING MODULE · USB-C / PLATFORM POWER · AI-ASSISTED HYPOTHESIS · REV B

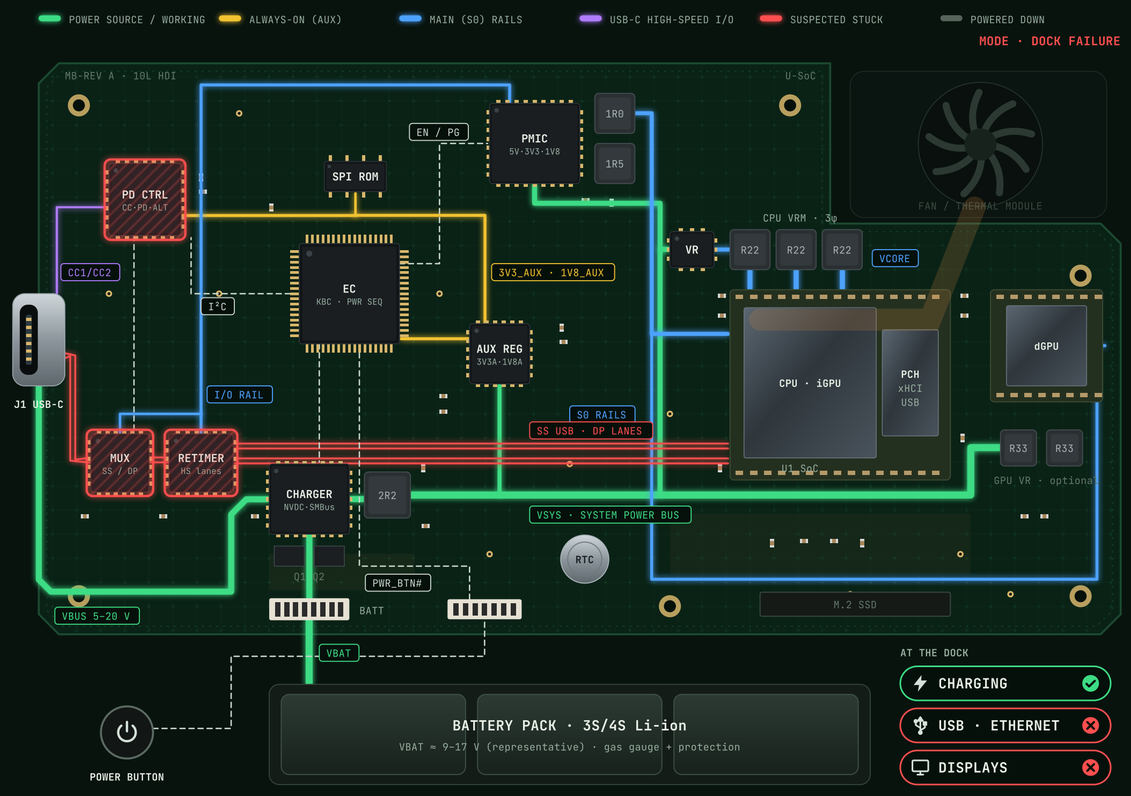
USB-C DOCK FAILURE EXPLAINED

HP USB-C Dock G5 → Windows business laptop · charging kept working · USB, Ethernet, displays gone — dock absent from Device Manager · restart did not help · ≈45 s hold fixed it

AI-assisted analysis · the explanation is a HYPOTHESIS, not a confirmed root cause · representative architecture, varies by model.



01 PCB MOTHERBOARD · WHERE THE STATE LIVES

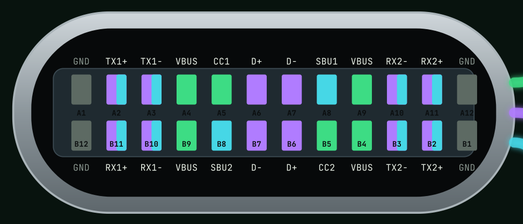


Representative modern business-laptop architecture — exact implementation varies by model. Red = suspected area (hypothesis).

02 USB-C SIGNAL PATH · ONE CONNECTOR, THREE JOBS

ONE CONNECTOR ≠ ONE SUBSYSTEM

USB-C receptacle, viewed face-on · 24 contacts · three independent jobs



- POWER · VBUS + CC (PD messages ride CC)
- USB 2.0 · D+/D-
- SUPERSPEED PAIRS — shared: USB 3 or DisplayPort, set by the mux
- SBU1/2 → DP AUX in Alt Mode
- GROUND

Plug is reversible: CC tells the controller which way round it is, then the mux maps TX1/RX1 or TX2/RX2 to the right destination.

POWER
VBUS · 5 / 9 / 15 / 20 V
CC1 / CC2 · attach + orientation
USB Power Delivery contract

USB DATA
D+ / D- · USB 2.0
SuperSpeed TX / RX pairs · USB 3
Dock hub → USB ports + Ethernet

DISPLAYPORT
High-speed lanes (2 or 4, via mux)
SBU1 / SBU2 → DP AUX channel
DisplayPort Alt Mode

INCIDENT
POWER — WORKING
USB DATA — MISSING
DISPLAYPORT — MISSING

Charging shows VBUS + CC + PD were working.
It says nothing about the data and display paths.

Observed: power stayed up; not even USB 2.0 enumerated. Hypothesis: the stuck step sat before data AND display, most likely the USB connection / data-role state the PD controller reports.

03 VOLTAGE DECAY · WHEN STATE IS ACTUALLY LOST



$V(t) = V_0 \cdot e^{(-t/RC)}$ · state survives until the rail crosses the retention threshold

04 WARM VS HARDWARE RESET

WINDOWS RESTART

warm reset · OS-controlled

- CPU → RESET — core reset
- WINDOWS → RELOAD — drivers re-init

BUT

- AUX POWER RAIL → STILL ON — never switched
- USB-C CONTROLLER → POWERED — keeps its RAM
- STUCK STATE → SURVIVES? — same bad state

RESULT: POWER ✓ · USB ✗ · DISPLAY ✗

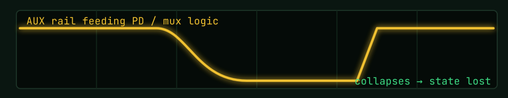


HARDWARE POWER-STATE RESET

long power-button hold · EC-controlled

- EC / POWER SEQUENCER — reads PWR_BTN#
- REGULATORS DISABLED — EN pins low
- RELEVANT RAILS COLLAPSE — caps discharge
- CONTROLLER LOSES STATE — volatile cleared
- COLD INITIALISATION — firmware reloads
- USB-C NEGOTIATES AGAIN — CC → PD → Alt Mode

RESULT: POWER ✓ · USB ✓ · DISPLAY ✓



PROPOSED MECHANISM · HYPOTHESIS

05 RESET TIMELINE · IS 45 s SPECIAL?

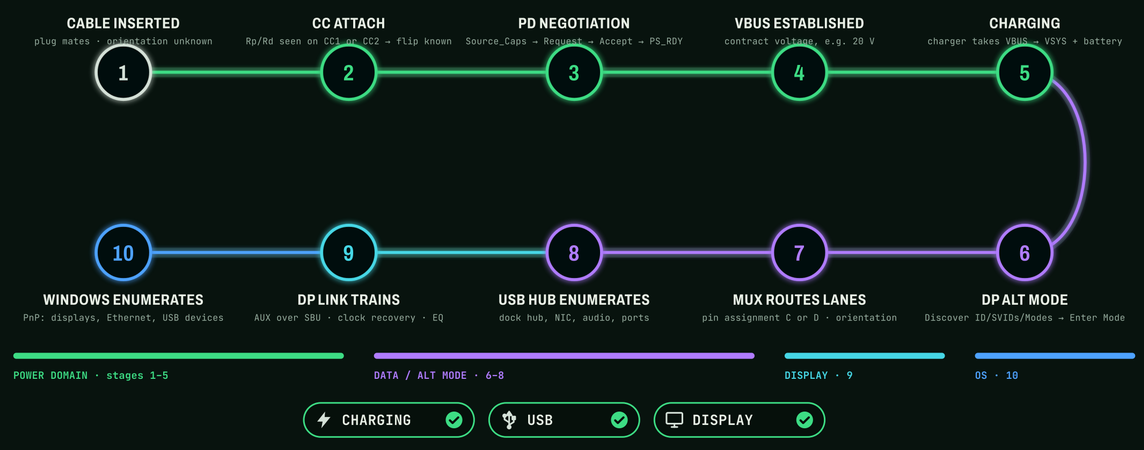


No documented procedure needs 45 s; all documented thresholds are ≤ 40 s. 45 s most likely = margin, not magic.

06 USB-C REINITIALISATION · CC → PD → ALT → USB → OS

NORMAL USB-C DOCK BRING-UP

Representative order · USB 2.0 can enumerate as soon as VBUS is up; DP and USB 3 wait for Alt Mode



Observed: after the reset every function returned, consistent with the chain re-running from stage 1.

07 DIAGNOSTIC DECISION PATH · THIS INCIDENT (AI-SUGGESTED)

DOCK CONNECTED → CHARGING? YES → USB WORKING? NO → DISPLAYS? MISSING → IN DEVICE MGR? NOT AT ALL → PRE-USB STATE? PD conn · EC → RESTART FIXED? NO → HARD RESET hold power → RETURNED? YES → TRANSIENT STATE strong hypothesis

Side exits: not charging → charger / cable / PD · USB OK → display path only · dock visible in Device Manager → Alt Mode / lane path · restart fixes it → driver / OS state · reset fails → hardware fault

Sources: ACPI 6.5 §4.8.2.2.1.3 · HP Support "Perform a power reset" · HP USB-C Dock G5 datasheet + SoftPaq firmware notes · Dell KB 000139016, 000125880 · Lenovo HT510479 · Microsoft Support 4036280